

Bayesian Foundations

Math 742

3/30/2026

Introduction

In Bayesian inference, unknown parameters are treated as random variables. Inference is based on probability statements about parameters conditional on observed data.

- **Frequentist:** parameters fixed, data random
- **Bayesian:** data observed, parameters random

Courtroom Analogy

Pretend you are a juror trying to decide if the defendant is guilty.

Bayesian Approach: What should I believe, given what I already know and the evidence I just saw?

- **Prior:** Your initial belief about guilt. The defendant is accused of speeding down main street. When you drive down main street everyone always seems to be speeding.
- **Likelihood:** How consistent the observed evidence is with being guilty or innocent. The police officer has radar evidence that the accused was driving 49mph, posted speed limit is 35mph.
- **Posterior:** $\text{Posterior} \propto \text{Prior} \times \text{Likelihood}$

Frequentist Approach: Suppose the defendant were innocent, how surprising would it be to see the evidence we saw.

- Prior beliefs are not explicitly modeled - based solely on the sampling distribution under H_0 .
- We assume **innocence** until proven guilty.
- We compute how extreme the evidence would be **if the person were innocent** (p-value).

Frequentist: Probability of data given the hypothesis
Bayesian: Probability of hypothesis given the data

Bayes' Theorem

Let X denote the observed data and θ an unknown parameter. Bayes' theorem gives

$$p(\theta | x) = \frac{p(x | \theta)\pi(\theta)}{p(x)}, \quad p(x) = \int p(x | \theta)\pi(\theta) d\theta.$$

- $\pi(\theta)$: prior distribution
- $p(x | \theta)$: likelihood
- $p(\theta | x)$: posterior distribution

- $p(x)$: marginal likelihood (normalizing constant)

Notice that we are often most interested in regions or points where the posterior density is high (e.g., the mode, also called the maximum a posteriori or MAP estimate). Because of this, the denominator $p(x)$ serves only as a normalizing constant to ensure the posterior integrates to 1. For most purposes we can work with the simpler unnormalized form:

$$p(\theta | x) \propto p(x | \theta) \pi(\theta)$$

The likelihood $p(x | \theta)$ is **not** a probability distribution in θ ; it is a function of θ for fixed data x . The prior $\pi(\theta)$ **is** a probability distribution, it encodes our beliefs about θ before seeing the data.

Bayesian updating is multiplicative. In words:

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}.$$

Bayesian inference is fundamentally about **updating prior beliefs** with data, not merely producing point estimates.

Example - Bernoulli:

Let $X | \theta \sim \text{Bernoulli}(\theta)$ with $x \in \{0, 1\}$.

The likelihood is

$$p(x | \theta) = \theta^x (1 - \theta)^{1-x}.$$

Assume a uniform prior on $(0, 1)$:

$$\pi(\theta) = 1, \quad 0 < \theta < 1.$$

Then the posterior satisfies

$$\pi(\theta | x) \propto p(x | \theta) \pi(\theta) = \theta^x (1 - \theta)^{1-x}.$$

- If $x = 1$, then $\pi(\theta | x) \propto \theta$
- If $x = 0$, then $\pi(\theta | x) \propto 1 - \theta$

Example: Suppose a manufacturer produces electronic components. Let,

θ = probability that a randomly selected component is defective.

We inspect **one component** and record, $X = 1$ if the component is defective or $X = 0$ if the component is not defective. Thus, $X \sim \text{Bernoulli}(\theta)$.

Using the likelihood and prior from above,

Likelihood

$$p(x|\theta) = \theta^x (1 - \theta)^{1-x}, 0 < \theta < 1$$

Prior

$$\pi(\theta) = 1, 0 < \theta < 1$$

And, we know that the posterior density is proportional to $\theta^x(1 - \theta)^{1-x}$ (i.e. not equal to).

We need to make sure this is indeed a probability density function in that it integrates to 1.

Case 1: $x = 1$

When $x = 1$ the posterior density is proportional to,

$$\theta^1(1 - \theta)^{1-1} = \theta$$

Here, θ is referred to as the *kernel*. The kernel is the part of the density that depends on θ , ignoring constants.

We need to find the normalizing constant so that the density integrates to 1.

$$C \int_0^1 \theta d\theta = 1$$

$$C \frac{\theta^2}{2} \Big|_0^1 = 1$$

$$\frac{C}{2} = 1 \implies C = 2$$

Case 2: $x = 0$

$$K \int_0^1 (1 - \theta) d\theta = 1$$

$$K \left(\theta - \frac{\theta^2}{2} \right) \Big|_0^1 = 1$$

$$K \left(1 - \frac{1}{2} \right) = 1$$

$$K = 2$$

The fact that both constants equal 2 is just a function of the fact that we have symmetry. The normalizing constants coincide here due to symmetry between success and failure under a uniform prior; they generally differ when the data are asymmetric or the prior is not symmetric.

Often, the normalizing constants will not be the same.

$$\pi(\theta \mid x = 1) = 2\theta, \quad 0 < \theta < 1$$

$$\pi(\theta \mid x = 0) = 2(1 - \theta), \quad 0 < \theta < 1$$

This posterior density happens to be a Beta probability density function with $\alpha = 2, \beta = 1$.

$$\text{Beta}(\alpha, \beta) : f(\theta \mid \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

Beta(α, β) with $\alpha > 1, \beta > 1$, mean = $\frac{\alpha}{\alpha + \beta}$ mode = $\frac{\alpha - 1}{\alpha + \beta - 2}$.

Suppose a failure (i.e. $x = 1$) is observed,

Based on the data, now what is our best guess of θ ? In other words, what is $\mathbb{E}(\theta|x = 1)$?

$$\mathbb{E}(\theta|x = 1) = \int_0^1 \theta 2\theta d\theta = 2 \frac{\theta^3}{3} \Big|_0^1 = \frac{2}{3}$$

The posterior mean $2/3$ is our updated expected value of the defect probability θ given we observed one defective item.

It is not the probability that $\theta > 0.5$. That requires integrating the posterior.

If we do not observe a failure, now what is our best guess of θ ?

$$\mathbb{E}(\theta|x = 0) = \int_0^1 \theta 2(1 - \theta) d\theta = 2 \left[\frac{\theta^2}{2} - \frac{\theta^3}{3} \right]_0^1 = 1 - 2/3 = 1/3$$

Posterior Interpretation

Common posterior summaries:

- Posterior mean
- Posterior median
- Posterior mode (MAP estimator)

A $100(1 - \alpha)\%$ credible interval $[a, b]$ satisfies

$$P(\theta \in [a, b] \mid x) = 1 - \alpha.$$

Interpretation:

There is $1 - \alpha$ posterior probability that θ lies in $[a, b]$.

Unlike a frequentist 95% confidence interval, which is a random interval that covers the fixed (unknown) θ 95% of the time in repeated sampling, the Bayesian credible interval directly states posterior probability content for θ given the observed data.

Follow-up Example (More Data: Binomial)

Suppose a manufacturer produces electronic components, and let

θ = probability that a randomly selected component is defective.

We inspect $n = 10$ components and observe $x = 7$ defectives. Let

$$X = \sum_{i=1}^{10} X_i,$$

where $X_i = 1$ if the i th component is defective and $X_i = 0$ otherwise. Then

Because the X_i are iid Bernoulli, their sum $X \sim \text{Binomial}(n = 10, p = \theta)$ with PMF:

$$p(x \mid \theta) = \binom{10}{x} \theta^x (1 - \theta)^{n-x}$$

Likelihood. The likelihood function is

$$p(x | \theta) = \binom{10}{7} \theta^7 (1 - \theta)^3 \propto \theta^7 (1 - \theta)^3, \quad 0 < \theta < 1.$$

Prior.

As before, we assume a uniform prior on $(0, 1)$:

$$\pi(\theta) = 1, \quad 0 < \theta < 1.$$

Posterior kernel.

By Bayes' rule, the posterior density satisfies

$$\pi(\theta | x) \propto p(x | \theta) \pi(\theta) \propto \theta^7 (1 - \theta)^3, \quad 0 < \theta < 1.$$

At this stage, we have identified the *kernel* of the posterior density, but we have not yet normalized it.

Before normalizing analytically, we approximate the posterior density numerically to visualize its shape. In more complex models, analytic normalization is often impossible, so simulation becomes essential.

- We sample from the prior, weight by the likelihood, and resample
- The red curve is the posterior we get numerically.
- The green line is the posterior mean, and the purple line is the MAP.
- This is Bayes' rule implemented by simulation.

```
set.seed(1)

# Data
n <- 10
x <- 7

# Prior: Uniform(0,1) = Beta(1,1)
alpha <- 1
beta <- 1

M <- 200000

# 1) Draw from prior
theta <- rbeta(M, alpha, beta)

# 2) Importance weights proportional to likelihood
w <- theta^x*(1-theta)^(n-x)
w <- w / sum(w)

# 3) Resample to approximate posterior draws
theta_post <- sample(theta, size = 50000, replace = TRUE, prob = w)

# Densities for plotting
d_prior <- density(theta, from = 0, to = 1)
d_post <- density(theta_post, from = 0, to = 1)

ymax <- max(d_prior$y, d_post$y)
```

```

plot(d_prior,
     lwd = 2,
     col = "dodgerblue",
     xlim = c(0, 1),
     ylim = c(0, ymax),
     main = "Prior vs Posterior (Monte Carlo Approximation)",
     xlab = expression(theta),
     ylab = "Density")

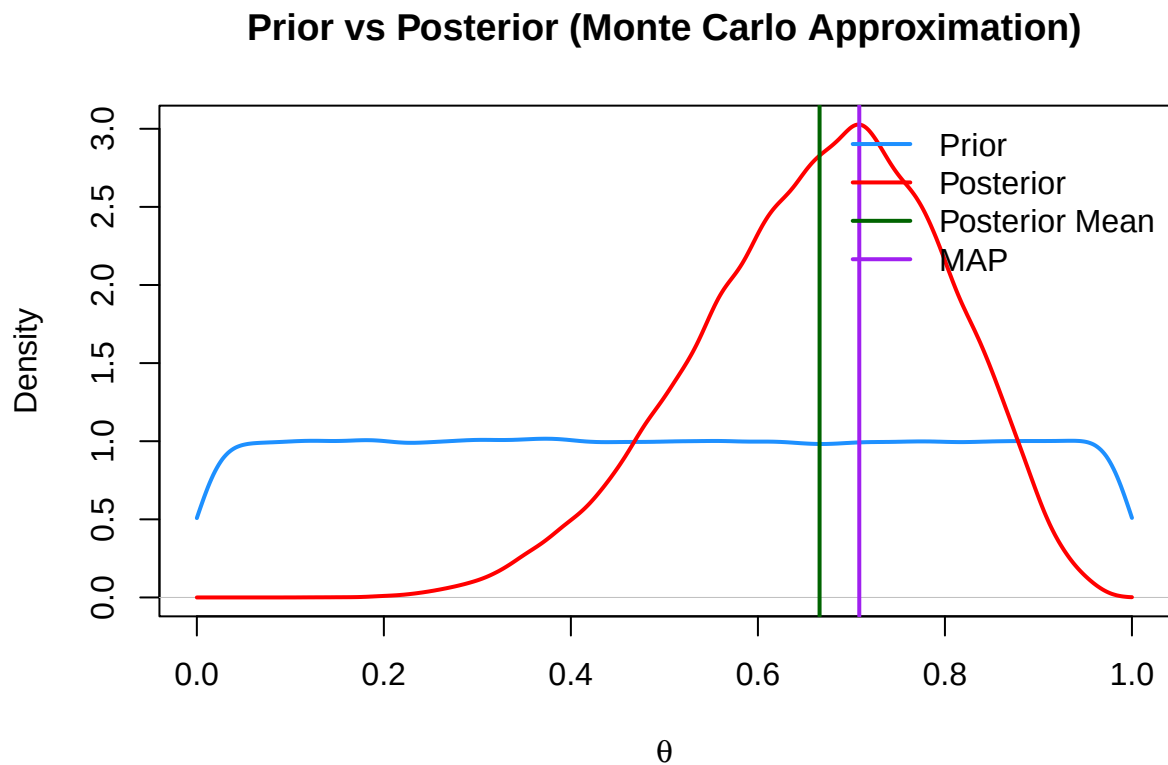
lines(d_post, lwd = 2, col = "red")

# Posterior summaries
post_mean <- mean(theta_post)
dmap <- density(theta_post, from = 0, to = 1)
post_map <- dmap$x[which.max(dmap$y)]

abline(v = post_mean, lwd = 2, col = "darkgreen")
abline(v = post_map, lwd = 2, col = "purple")

legend("topright",
     legend = c("Prior", "Posterior", "Posterior Mean", "MAP"),
     col = c("dodgerblue", "red", "darkgreen", "purple"),
     lwd = 2,
     bty = "n")

```



```
# Numerical summaries
post_mean

## [1] 0.6659333
quantile(theta_post, c(0.025, 0.975))

##      2.5%      97.5%
## 0.3840667 0.8901837
post_map

## [1] 0.7084149
```

Recognizing the posterior distribution

The posterior kernel $\theta^7(1 - \theta)^3$ corresponds to a Beta density.

Normalizing analytically, we obtain

$$\theta \mid x \sim \text{Beta}(8, 4).$$

Posterior Mean

The posterior mean is

$$\mathbb{E}(\theta \mid x) = \int_0^1 \theta \pi(\theta \mid x) d\theta.$$

Since $\theta \mid x \sim \text{Beta}(8, 4)$, we have

$$\mathbb{E}(\theta \mid x) = \frac{8}{8 + 4} = \frac{2}{3}.$$

The posterior mean lies between the prior mean (1/2) and the sample proportion (7/10), illustrating shrinkage toward the prior.

Interpretation.

The posterior mean is a compromise between the prior mean (1/2) and the sample proportion (7/10), with the data pulling the estimate toward 0.7.

Most Likely Value of θ (MAP Estimate)

In addition to the posterior mean, another common Bayesian point estimate is the *maximum a posteriori* (MAP) estimate. The MAP estimate is the value of θ that maximizes the posterior density.

Recall that the posterior density is

$$\pi(\theta \mid x) \propto \theta^7(1 - \theta)^3, \quad 0 < \theta < 1.$$

Thus, the MAP estimate is the value of θ that maximizes

$$f(\theta) = \theta^7(1 - \theta)^3.$$

Because the logarithm is strictly increasing so it preserves maximizers. Thus, instead maximize the log-posterior:

$$\ell(\theta) = \log f(\theta) = 7 \log \theta + 3 \log(1 - \theta).$$

Taking the derivative with respect to θ ,

$$\ell'(\theta) = \frac{7}{\theta} - \frac{3}{1 - \theta}.$$

Setting the derivative equal to zero,

$$\frac{7}{\theta} = \frac{3}{1 - \theta}.$$

Solving for θ ,

$$7(1 - \theta) = 3\theta \quad \Rightarrow \quad 7 = 10\theta \quad \Rightarrow \quad \theta = 0.7.$$

Since the posterior density is zero at $\theta = 0$ and $\theta = 1$ and positive for $0 < \theta < 1$, this critical point corresponds to a maximum.

Conclusion:

$$\theta_{\text{MAP}} = 0.7.$$

Remark: With a uniform prior, the MAP estimate coincides with the MLE $x/n = 7/10$.

How the MAP Changes Under a Non-Uniform Beta Prior

To highlight the influence of the prior, suppose we use a Beta prior

$$\theta \sim \text{Beta}(\alpha, \beta), \quad \alpha > 0, \beta > 0,$$

so that

$$\pi(\theta) \propto \theta^{\alpha-1}(1 - \theta)^{\beta-1}, \quad 0 < \theta < 1.$$

We again observe $x = 7$ defectives out of $n = 10$ inspected items, so the likelihood is

$$p(x | \theta) \propto \theta^7(1 - \theta)^3.$$

Posterior

Multiplying prior and likelihood gives

$$\pi(\theta | x) \propto \theta^7(1 - \theta)^3 \cdot \theta^{\alpha-1}(1 - \theta)^{\beta-1} = \theta^{(\alpha+7)-1}(1 - \theta)^{(\beta+3)-1}.$$

Hence

$$\theta | x \sim \text{Beta}(\alpha + 7, \beta + 3)$$

MAP estimate

The MAP is the maximizer of the posterior density. For a $\text{Beta}(a, b)$ distribution with $a > 1$ and $b > 1$, the mode is

$$\theta_{\text{MAP}} = \frac{a - 1}{a + b - 2}.$$

Applying this to the posterior with $a = \alpha + 7$ and $b = \beta + 3$ (and assuming $\alpha + 7 > 1$ and $\beta + 3 > 1$), we obtain

$$\theta_{\text{MAP}} = \frac{(\alpha + 7) - 1}{(\alpha + 7) + (\beta + 3) - 2} = \frac{\alpha + 6}{\alpha + \beta + 8}.$$

Interpretation

When the prior is uniform, $\alpha = \beta = 1$, so

$$\theta_{\text{MAP}} = \frac{1 + 6}{1 + 1 + 8} = \frac{7}{10} = 0.7,$$

which matches the MLE x/n .

With a non-uniform prior, the MAP shifts toward the prior mode.

Example A (Prior favors smaller defect rates). Suppose we choose $\theta \sim \text{Beta}(2, 8)$, which places more prior mass near small values of θ . Then

$$\theta_{\text{MAP}} = \frac{\alpha + 6}{\alpha + \beta + 8} = \frac{2 + 6}{2 + 8 + 8} = \frac{8}{18} \approx 0.444.$$

Even though the sample proportion is $7/10 = 0.7$, the MAP estimate is pulled downward by the prior.

Example B (Prior favors larger defect rates).

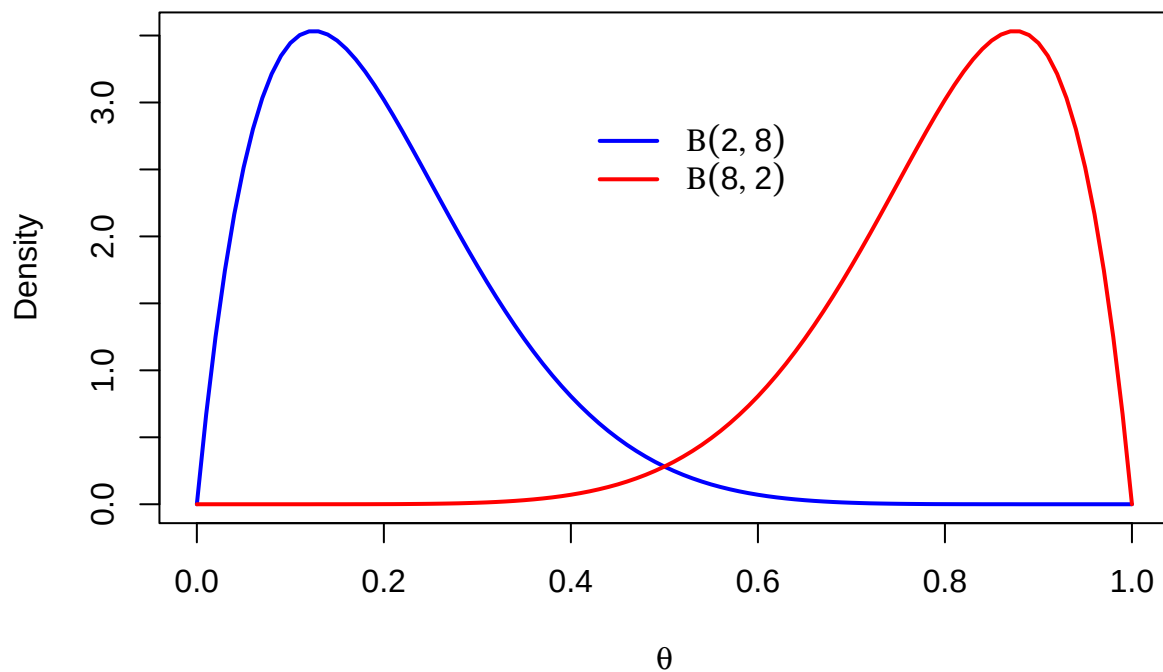
Suppose instead $\theta \sim \text{Beta}(8, 2)$, which places more prior mass near large values of θ . Then

$$\theta_{\text{MAP}} = \frac{8 + 6}{8 + 2 + 8} = \frac{14}{18} \approx 0.778,$$

so the MAP estimate is pulled upward.

Takeaway

The MAP estimate balances the data and the prior. A stronger prior (larger $\alpha + \beta$) exerts more influence and can noticeably shift the MAP away from the MLE.



- Blue: prior believes θ is small.
- Red: prior believes θ is large.

Summary

- Bayesian inference follows directly from Bayes' Theorem
- Prior beliefs and data are combined multiplicatively
- Interpretation is probabilistic and direct

Next: Conjugate priors give closed-form posteriors.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Seed (matches set.seed(1) as closely as possible)
np.random.seed(1)

# Data
n = 10
x = 7

# Prior: Uniform(0,1) = Beta(1,1)
alpha = 1
beta = 1
```

```

M = 200000

# 1) Draw from prior
theta = np.random.beta(alpha, beta, size=M)

# 2) Importance weights proportional to likelihood
logw = x * np.log(theta) + (n - x) * np.log(1.0 - theta)
w = np.exp(logw - np.max(logw)) # stabilize
w = w / np.sum(w)

# 3) Resample to approximate posterior draws
theta_post = np.random.choice(theta, size=50000, replace=True, p=w)

# Helper: simple Gaussian KDE (no external deps)
def kde_gaussian(samples, grid, bandwidth=None):
    samples = np.asarray(samples)
    n_samp = samples.size
    if bandwidth is None:
        # Silverman's rule of thumb
        std = np.std(samples, ddof=1)
        bw = 1.06 * std * (n_samp ** (-1/5))
        bw = max(bw, 1e-6)
    else:
        bw = float(bandwidth)

    u = (grid[:, None] - samples[None, :]) / bw
    dens = np.mean(np.exp(-0.5 * u * u), axis=1) / (bw * np.sqrt(2.0 * np.pi))
    return dens

# Densities for plotting
grid = np.linspace(0.0, 1.0, 401)
d_prior = kde_gaussian(theta, grid)
d_post = kde_gaussian(theta_post, grid)

ymax = max(np.max(d_prior), np.max(d_post))

# Plot
plt.figure()
plt.plot(grid, d_prior, linewidth=2, color="dodgerblue", label="Prior")
plt.plot(grid, d_post, linewidth=2, color="red", label="Posterior")
plt.xlim(0, 1)

## (0.0, 1.0)
plt.ylim(0, ymax)

## (0.0, 2.912174346272392)
plt.title("Prior vs Posterior (Monte Carlo Approximation)")
plt.xlabel(r"$\theta$")
plt.ylabel("Density")

# Posterior summaries
post_mean = float(np.mean(theta_post))

```

```

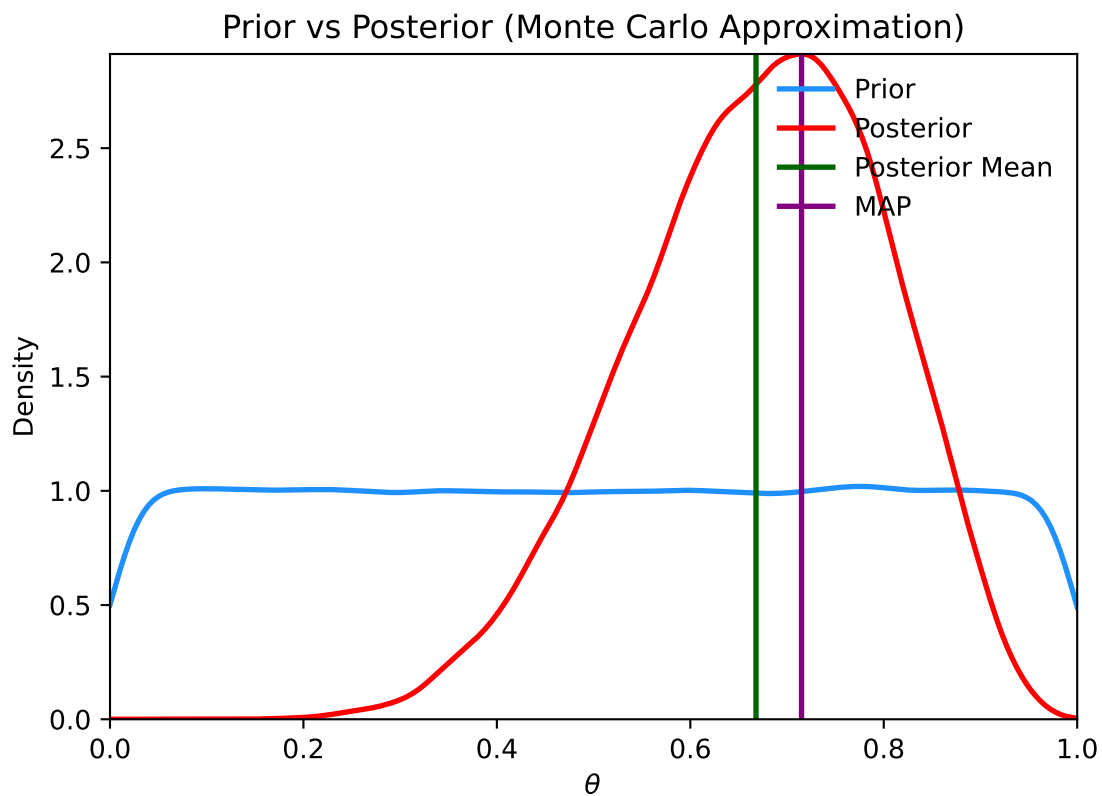
post_ci = np.quantile(theta_post, [0.025, 0.975])

dmap = kde_gaussian(theta_post, grid)
post_map = float(grid[np.argmax(dmap)])

# Vertical lines
plt.axvline(post_mean, linewidth=2, color="darkgreen", label="Posterior Mean")
plt.axvline(post_map, linewidth=2, color="purple", label="MAP")

plt.legend(loc="upper right", frameon=False)
plt.show()

```



```

# Numerical summaries
post_mean

## 0.6679111541570202

post_ci

## array([0.39268949, 0.89195091])

post_map

## 0.715

```